

Project Planning Report

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EcoRoute Campus Shuttle Optimization and Real-Time Tracking System

Executive Summary

This report includes a full project planning report on project initiation, SDLC strategy, WBS, PERT scheduling, Intermediate COCOMO estimation, risk assessment, feature prioritization and triple constraint balancing.

Therefore, the software project on EcoRoute Campus Shuttle Optimization and Real-Time Tracking System aims to optimize the use of campus shuttles and provide real-time tracking data for university students, drivers, and transport administrators. It offers live tracking of shuttle locations, estimated arrival times, routing assistance, driver trip statuses and an admin dashboard for managing transport operations.

The PERT expected project duration is 40.00 weeks. The critical path is A -> B -> D -> H -> I -> J -> K. Under the standard PERT normal approximation, the probability of completion is 50.00% since the target completion date equals the PERT expected duration. An Intermediate COCOMO estimate for a 10 KLOC project that is semidetached, yields an effort of 44.21 person-months and a development time of 9.42 months.

1. Project Initiation and Lifecycle Strategy

1.1 Project Title

EcoRoute Campus Shuttle Optimization and Real-Time Tracking System

1.2 Project Scope

The project will involve the creation of a web and mobile software system to track the movement of the campus shuttles, improve routing and provide accurate shuttle arrival information to students. To clearly define project boundaries, the scope is broken down into In-Scope items and Out-of-Scope items.

In-Scope Items

- Student mobile application for real-time shuttle location and arrival time.
- Mobile module for driver to share GPS, trip start/end status, and report route updates.
- Admin Web Dashboard to manage shuttle routes, stops, schedules, drivers and reports.
- Backend API and database to store user data, shuttle data, GPS logs, routes and notifications.
- Route optimization system to recommend efficient routes for campus shuttle based on the demand and stop information.
- Delay, route change and shuttle arrival push notification service.
- Testing, pilot deployment and user training of selected campus environment.

Out-of-Scope Items

- The purchase of physical shuttle vehicles or GPS hardware devices.
- Self-driving (or autonomous) control of shuttle.
- Payment of fares or financial transactions via the Internet.
- Fully connected with city-wide public transportation systems.
- Advanced traffic signal control/ government road infrastructure management.
- Multiple universities has been supported in the initial release.

1.3 SMART Objectives

ID	Objective Type	SMART Objective
O1	Operational Efficiency	Complete and deploy a working shuttle tracking MVP within 40 weeks, including student app, driver GPS module, admin dashboard, and backend API.
O2	Performance	Provide shuttle location updates within a maximum average delay of 10 seconds during pilot testing.
O3	User Adoption	Achieve at least 70% pilot-user satisfaction among 100 student testers and 5 transport staff by the end of user acceptance testing.

1.4 SDLC Model Selection and Justification

In this project, the selected model of SDLC is called Agile Incremental Model. The modules in the project are multiple user-facing modules which include the student application, driver module, admin dashboard, notifications, and route optimization. These modules can be built and tested in small increments in the form of sprints. The use of Agile is appropriate because transport office staff and students can give feedback on the use of early prototypes, and the team can make improvements to the features without having to wait until the entire system is completed.

Agile is more suitable for this project, compared to the Waterfall Model. Once requirements are stable at the start then changes to the requirement are costly in Waterfall. Route policies, stop locations, notification rules and dashboard reports can be modified following pilot feedback in a campus shuttle system. Agile encourages incremental development, early testing, customer feedback, and mitigates risk with frequent releases.

2. Scope Decomposition and Scheduling (PERT)

2.1 Work Breakdown Structure (WBS)

The following text-based WBS sketch shows the project decomposition into three levels with more than 12 work packages.

1. EcoRoute Campus Shuttle System

1.1 Project Management:

1.1.1 Stakeholder interviews;

1.1.2 Project charter and scope baseline;

1.1.3 Sprint planning and progress monitoring

1.2 Requirements and Design:

1.2.1 Functional requirement specification;

1.2.2 Non-functional requirement specification;

1.2.3 UI/UX wireframes and prototype;

1.2.4 Database design;

1.2.5 API and system architecture design

1.3 Software Development:

1.3.1 Student mobile app;

1.3.2 Driver GPS module;

1.3.3 Admin web dashboard;

1.3.4 Route optimization engine;

1.3.5 Notification service

1.4 Testing and Deployment:

1.4.1 Unit testing;

1.4.2 Integration and system testing;

1.4.3 Security and privacy testing;

1.4.4 Pilot deployment;

1.4.5 User training and final release

2.2 PERT Activity Table

The activity estimates are given in weeks. PERT uses three estimates for each activity: Optimistic (O), Most Likely (M), and Pessimistic (P).

ID	Activity	Predecessor	O	M	P	TE	Variance
A	Requirement Analysis	None	2	4	6	4.00	0.44
B	System Architecture and SDLC Planning	A	2	4	8	4.33	1.00
C	UI/UX Prototyping	A	2	4	6	4.00	0.44
D	Database and API Design	B	2	4	8	4.33	1.00
E	Student Mobile App Development	C, D	6	10	16	10.33	2.78
F	Driver GPS Module Development	D	4	8	14	8.33	2.78
G	Admin Dashboard Development	C, D	4	6	12	6.67	1.78
H	Route Optimization Engine	D	6	10	18	10.67	4.00
I	Integration and System Testing	E, F, G, H	4	8	14	8.33	2.78
J	Pilot Deployment and User Training	I	2	4	8	4.33	1.00
K	User Acceptance and Final Release	J	2	4	6	4.00	0.44

2.3 PERT Formulas and Sample Calculation

Expected Time (TE) = $(O + 4M + P) / 6$

Variance = $[(P - O) / 6]^2$

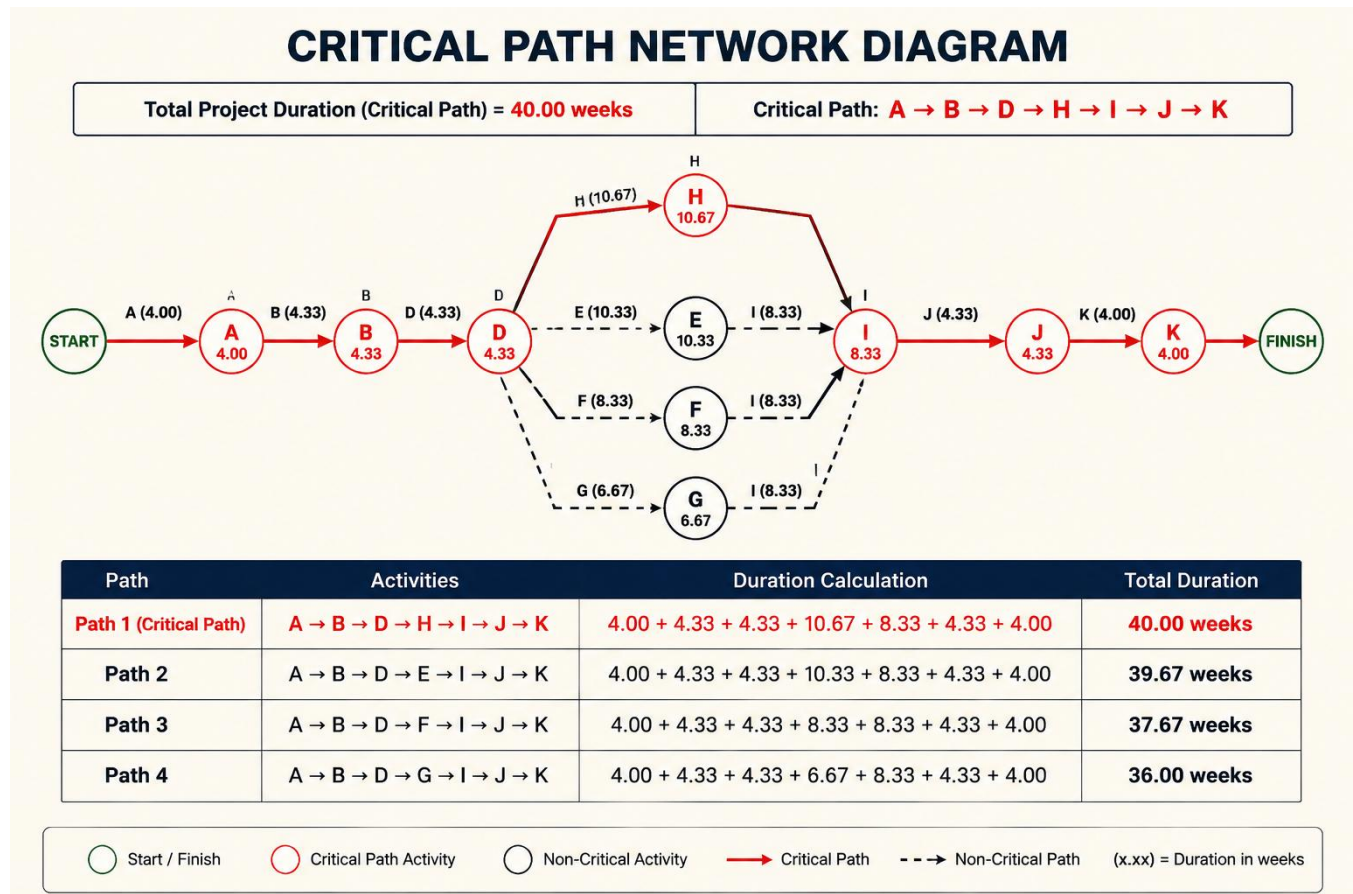
Sample calculation for Activity A - Requirement Analysis:

TE_A = $(2 + 4(4) + 6) / 6 = 24 / 6 = 4.00$ weeks

Variance_A = $[(6 - 2) / 6]^2 = (4 / 6)^2 = 0.44$

2.4 Critical Path Determination

The main possible paths and their durations are calculated below:



Therefore, the Critical Path is A -> B -> D -> H -> I -> J -> K, and the total expected project duration is 40.00 weeks.

Project Activity Table

ID	Activity	Predecessor	TE (weeks)	ES	EF	LS	LF	Total Float	Free Float	Critical Path?
A	Requirement Analysis	None	4.00	0	4.00	0	4.00	0	0	Yes
B	System Architecture & SDLC Planning	A	4.33	4.00	8.33	4.00	8.33	0	0	Yes
D	Database & API Design	B	4.33	8.33	12.66	8.33	12.66	0	0	Yes
H	Route Optimization Engine	D	10.67	12.66	23.33	12.66	23.33	0	0	Yes
I	Integration & System Testing	E,F,G,H	8.33	23.33	31.66	23.33	31.66	0	0	Yes
J	Pilot Deployment & User Training	I	4.33	31.66	35.99	31.66	35.99	0	0	Yes
K	User Acceptance & Final Release	J	4.00	35.99	39.99	35.99	39.99	0	0	Yes

2.5 Probability of Completion

To calculate the completion probability, the variance values of the critical path activities are added first.

$$\text{Critical Path Variance} = \text{Var}(A) + \text{Var}(B) + \text{Var}(D) + \text{Var}(H) + \text{Var}(I) + \text{Var}(J) + \text{Var}(K)$$

$$\text{Critical Path Variance} = 0.44 + 1.00 + 1.00 + 4.00 + 2.78 + 1.00 + 0.44 = 10.67$$

$$\text{Standard Deviation, } \sigma = \sqrt{10.67} = 3.27 \text{ weeks}$$

$$Z = (D - TE) / \sigma$$

Here, the required probability is for completing the project on or before the estimated duration.

Therefore, $D = TE = 40.00$ weeks.

$$Z = (40.00 - 40.00) / 3.27 = 0.00$$

$$P(Z \leq 0.00) = 0.5000 = 50.00\%$$

Therefore, the statistical probability of completing the project on or before the expected duration of 40.00 weeks is 50.00% under the standard PERT normal approximation. This is reasonable because the target date is exactly equal to the expected mean duration. If a management buffer target of 44 weeks is used, $Z = (44 - 40) / 3.27 = 1.22$ and the probability becomes approximately 88.97%.

3. Effort and Cost Estimation (Intermediate COCOMO)

3.1 Size and Development Mode

The estimated software size is 10 KLOC, meaning approximately 10,000 delivered source lines of code. This estimate is based on a compact MVP using existing mobile and web frameworks, a backend API, database layer, GPS integration, notification service, and a moderate route optimization component. Third-party mapping services reduce the need to build map rendering from scratch.

The selected COCOMO mode is Semidetached. The project is not extremely small or fully routine because it involves mobile apps, live GPS data, backend services, route optimization, and multiple user groups. However, it is also not a strict embedded safety-critical system. Therefore, Semidetached mode is appropriate.

3.2 Cost Drivers and EAF

Driver	Cost Driver	Rating	Multiplier	Justification
RELY	Required Software Reliability	High	1.15	Live shuttle information must be dependable for students and transport staff.
DATA	Database Size	High	1.08	The system stores users, vehicles, GPS logs, routes, schedules, and trip history.
CPLX	Product Complexity	High	1.15	The project includes GPS tracking, notifications, dashboards, and route optimization.
ACAP	Analyst Capability	High	0.86	A capable analyst reduces rework through better requirements and process modeling.
TOOL	Use of Software Tools	High	0.91	Frameworks, mapping APIs, CI tools, and testing tools improve productivity.
SCED	Required Development Schedule	Nominal	1.00	The planned schedule is demanding but not extreme.

$EAF = RELY \times DATA \times CPLX \times ACAP \times TOOL \times SCED$

$EAF = 1.15 \times 1.08 \times 1.15 \times 0.86 \times 0.91 \times 1.00 = 1.1178 \approx 1.12$

3.3 Effort and Development Time Calculation

Intermediate COCOMO effort equation for Semidetached mode:

$$\text{Effort} = a \times (\text{KLOC})^b \times \text{EAF}$$

For Semidetached mode: $a = 3.0$ and $b = 1.12$

$$\text{Effort} = 3.0 \times (10)^{1.12} \times 1.1178$$

$$\text{Effort} = 44.21 \text{ Person-Months}$$

Development time equation:

$$\text{TDEV} = c \times (\text{Effort})^d$$

For Semidetached mode: $c = 2.5$ and $d = 0.35$

$$\text{TDEV} = 2.5 \times (44.21)^{0.35}$$

$$\text{TDEV} = 9.42 \text{ months}$$

$$\text{Average Staffing} = \text{Effort} / \text{TDEV} = 44.21 / 9.42 = 4.69 \approx 5 \text{ people}$$

4. Risk Assessment and Mitigation

4.1 Risk Matrix and Risk Exposure

Risk Exposure (RE) is calculated using the formula: $\text{RE} = \text{Likelihood} \times \text{Impact}$. Likelihood is scored from 0.1 to 1.0, and impact is scored from 1 to 10.

ID	Risk	Likelihood	Impact	RE	Explanation
R1	GPS or map API service outage	0.50	9	4.50	Live location tracking may temporarily fail.
R2	Route optimization gives inaccurate suggestions	0.60	8	4.80	Inefficient routes may reduce trust in the system.
R3	Low student adoption during pilot	0.40	7	2.80	Students may continue using informal communication channels.
R4	Security or privacy breach of location data	0.30	10	3.00	Unauthorized access to user or vehicle location data would be serious.
R5	Schedule delay in mobile app development	0.70	8	5.60	Mobile work may take longer because of device testing and GPS issues.
R6	Scope creep from transport office requests	0.60	9	6.30	Additional reports, routes, or policy changes may expand scope.

4.2 Prioritized Mitigation and Contingency Plans

Priority Risk	Mitigation Plan	Contingency Plan
R5 - Schedule delay in mobile app development	Use sprint planning, early GPS prototype, weekly progress review, reusable UI components, and continuous integration testing.	Add one mobile developer temporarily, reduce non-essential screens, and prioritize core tracking and notification features first.
R6 - Scope creep from transport office requests	Freeze MVP scope after requirement approval and maintain a formal change request log with impact analysis.	Move new non-critical requests to version 2.0 and protect the critical path tasks from uncontrolled changes.
R2 - Inaccurate route optimization	Validate the algorithm using historical shuttle data, driver feedback, and simulated demand scenarios before deployment.	Use manually approved admin route rules until the optimization model is corrected and retested.

5. Feature Prioritization

5.1 Feature Options

- F1: Real-time Shuttle Tracking - students can view shuttle positions and basic arrival information.
- F2: Predictive Arrival Notifications - students receive alerts for shuttle arrival, delays, and route changes.
- F3: AI-Based Route Optimization - the system suggests more efficient routes using stop demand and shuttle movement data.

5.2 Weighted Scoring Model

The scoring scale is 1 to 5, where 5 is the best. The criteria weights are: Business Value = 40%, Feasibility = 30%, and Cost Efficiency = 30%. For Cost Efficiency, a higher score means lower relative cost and better value for the effort.

ID	Feature	Business Value	Feasibility	Cost Efficiency	Weighted Score
F1	Real-time Shuttle Tracking	5	5	4	$(5 \times 0.40) + (5 \times 0.30) + (4 \times 0.30) = 4.70$
F2	Predictive Arrival Notifications	4	4	5	$(4 \times 0.40) + (4 \times 0.30) + (5 \times 0.30) = 4.30$
F3	AI-Based Route Optimization	5	3	3	$(5 \times 0.40) + (3 \times 0.30) + (3 \times 0.30) = 3.80$

The highest-scoring feature is F1 - Real-time Shuttle Tracking, with a weighted score of 4.70. Therefore, it should be selected for priority development because it delivers immediate value to students, is technically feasible, and supports the rest of the system.

6. Conclusion and Triple Constraint Balancing

6.1 Synthesis of Key Project Planning Metrics

Metric	Value
Project Title	EcoRoute Campus Shuttle Optimization and Real-Time Tracking System
Selected SDLC Model	Agile Incremental Model
Estimated Size	10 KLOC
COCOMO Mode	Semidetached
Effort Adjustment Factor	1.12
COCOMO Effort	44.21 Person-Months
COCOMO Development Time	9.42 months
PERT Expected Duration	40.00 weeks
Critical Path	A -> B -> D -> H -> I -> J -> K
Critical Path Standard Deviation	3.27 weeks
Probability by Expected Duration	50.00%
Highest Priority Feature	Real-time Shuttle Tracking
Top Risks	Schedule delay, scope creep, inaccurate route optimization

6.2 Triple Constraint Trade-off Analysis

The 3 constraints are Scope, Time and Cost. There are 40 weeks of project work which are currently anticipated. If the client requests the schedule reduction of 20%, the revised schedule duration will be:
Schedule Reduction = $40 \times 20\% = 8$ weeks

New Target Schedule = $40 - 8 = 32$ weeks

A 32 week target is ambitious as it takes 8 weeks off the project time. The project manager should not compromise the cost, schedule and scope by simply reducing testing, since this would raise the quality and safety risks. A controlled rebalance should be used:

- Scope adjustment: Continue with the core real-time tracking, driver GPS module and the admin route management in the MVP, and push some analytics reports and advanced route optimisation through AI for later.
- Cost adjustment: Provide temporary mobile and testing resources to activities that are critical path components, particularly those activities for route engine development, integration testing, and pilot preparation.
- Time adjustment: Parallel development of UI, backend, and GPS module; as long as there are dependencies, and use pre-built mapping and notification APIs to cut in development time.
- Quality protection: Continue integration testing, security testing, and pilot validation; reduce low priority features, not essential quality testing.

So, if the schedule is cut 20%, the optimal solution is cut back on nonessential scope, increase development cost short term, while maintaining quality. In the next increment, the MVP should concentrate on the highest priority feature – real-time tracking of the shuttle, and then the more advanced optimization features.

References

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- Pressman, R. S., & Maxim, B. R. Software Engineering: A Practitioner's Approach.
- Project Management Institute. A Guide to the Project Management Body of Knowledge (PMBOK Guide).